

int x = -5;

int y = 3;

$y = y + x +$
 $y = 3 + (-5) + 1$

S.o.pln (x + " + y);

y = y +

-6

-26/3

-6

-9

-5

x = -6

-7

-6

x - - + - - x + + + x;

$3 + (-5) + 1 - (-5) + (-7)$
 $3 - 5 + 1 + 5 - 7 - 6$
 $+ (-6)$

-5

-3

-6

-6

Type Casting

Converting one Data type to other.

Two ways

① Widening / Automatic / implicit

byte $\rightarrow$ short $\rightarrow$ char $\rightarrow$ int $\rightarrow$ long $\rightarrow$ float $\rightarrow$ double

② Narrowing / Manual / Explicit

double $\rightarrow$ float $\rightarrow$ long $\rightarrow$ int $\rightarrow$ char $\rightarrow$ short $\rightarrow$ byte

① widening / implicit

```
int a = 5;  
double d = a; ✓  
System.out.println(d);  $\rightarrow$  5.0
```

② Narrowing / Explicit

```
double d = 5.2;  
int a = d;   
System.out.println(a);
```

error.

5

Syntax

Destⁿ Variable = (Destⁿ Variable Data type) Variable / Value / exp.ⁿ ;

```
int a = (int) d;  
5  $\uparrow$  (int) 5.2  
5
```

double a = 3.0;

int c = (8/a + 5.1); → int c = (int)(8/a + 5.1);

s.o.pln(c);

$$8/3.0 + 5.1$$

$$= 2.6 + 5.1$$

$$7.7$$

int c = $\left(\underbrace{8/(int)a}_{8/3 + 5} + (int)5.1 \right)$

⑦ = 2 + 5

6. (B)

double a = 3.14159;

scanf("%lf", &a); $\longrightarrow$ 3.14159

scanf("%lf", &a+1); $\longrightarrow$ 4.14159

scanf("%lf", &(int)a); $\longrightarrow$ $8 / (\text{int}) 3.14159 = 8/3 = 2$

scanf("%lf", &a); $\longrightarrow$ $8 / 3.14159 = 2.546$

scanf("%lf", &(int)(8/a)); $\longrightarrow$ $(\text{int})(8/3.14159) = (\text{int})2.546 = 2$

IPS - 1 ^{GP}
(6)

Attendance

5

Midterm

15

Quiz

20

EndSem

60

Operators

Precedence Level

Associativity Rule.

↑ H

L

ASCII Value.

American standard Code for int^m & interchange.

Char c;

int a = 65;

1. c = a;

s.o.pln(c);

Char g = 'F';

A

int ↔ char

Compatible to each other.

Numer/char

ASCII Value

A → 65

B → 66

⋮

Z → 90

a → 97

b → 98

⋮

Z → 122

0 → 48

1 → 49

⋮

9 → 57

Space bar
32

Char c = ' ';
int a = c;
s.o.pln(a);
↓
32

Char c = 'F';
int a = c;
s.o.pln(a); → 70

Enter 1st 4 character of Ur 1st name and Print sum of its ASCII Value.

String $S = \text{"SUNI"}$
 Storing $S_1 = \text{"PUMA"}$
 find sum
 S

$P = 80$	$S \rightarrow 83$
$U = 85$	$U \rightarrow 85$
$M = 77$	$N \rightarrow 78$
$A = 65$	$I \rightarrow 73$
<u>Sum = 307</u>	<u>Sum = 319</u>

String S ;
 Scanner $sc = \dots$
 $sc.println(\text{"Enter"});$
 $S = sc.next();$
 int a , $sum = 0$

83 $a = S.charAt(0);$

$sum = sum + a;$
 83 0 + 83

85 $a = S.charAt(1);$

$sum = sum + a$
 168 : 83 + 85

78 $a = S.charAt(2);$

$sum = sum + a$
 246 : 168 + 78

$S = \text{"SUNI"}$
 index $\rightarrow$

S	U	N	I
0	1	2	3

String Variable. charAt(index)

$S.charAt(0) \rightarrow S$

$S.charAt(1) \rightarrow U$

73 $a = S.charAt(3);$
 I

$sum = sum + a$
 319 = 246 + 73